

UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

Last updated 2/10/25

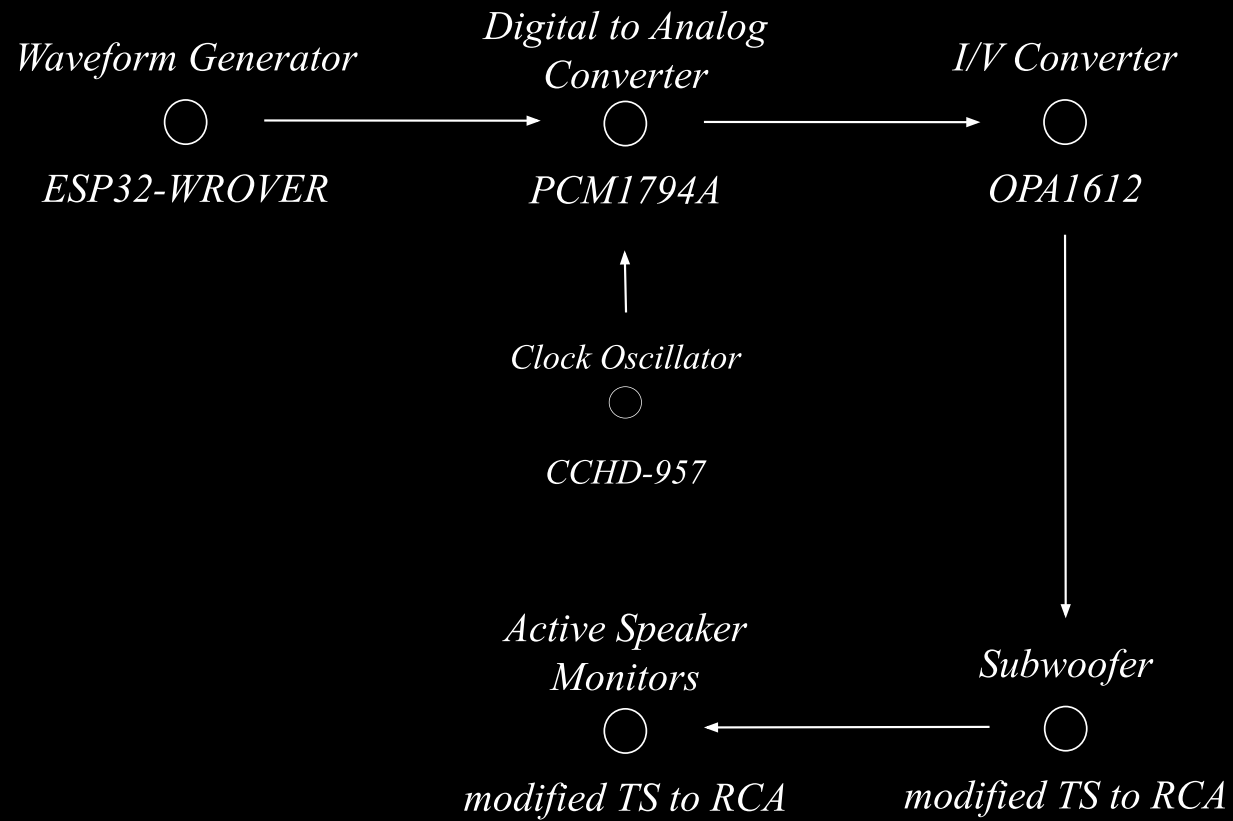
PROJECT OBJECTIVES

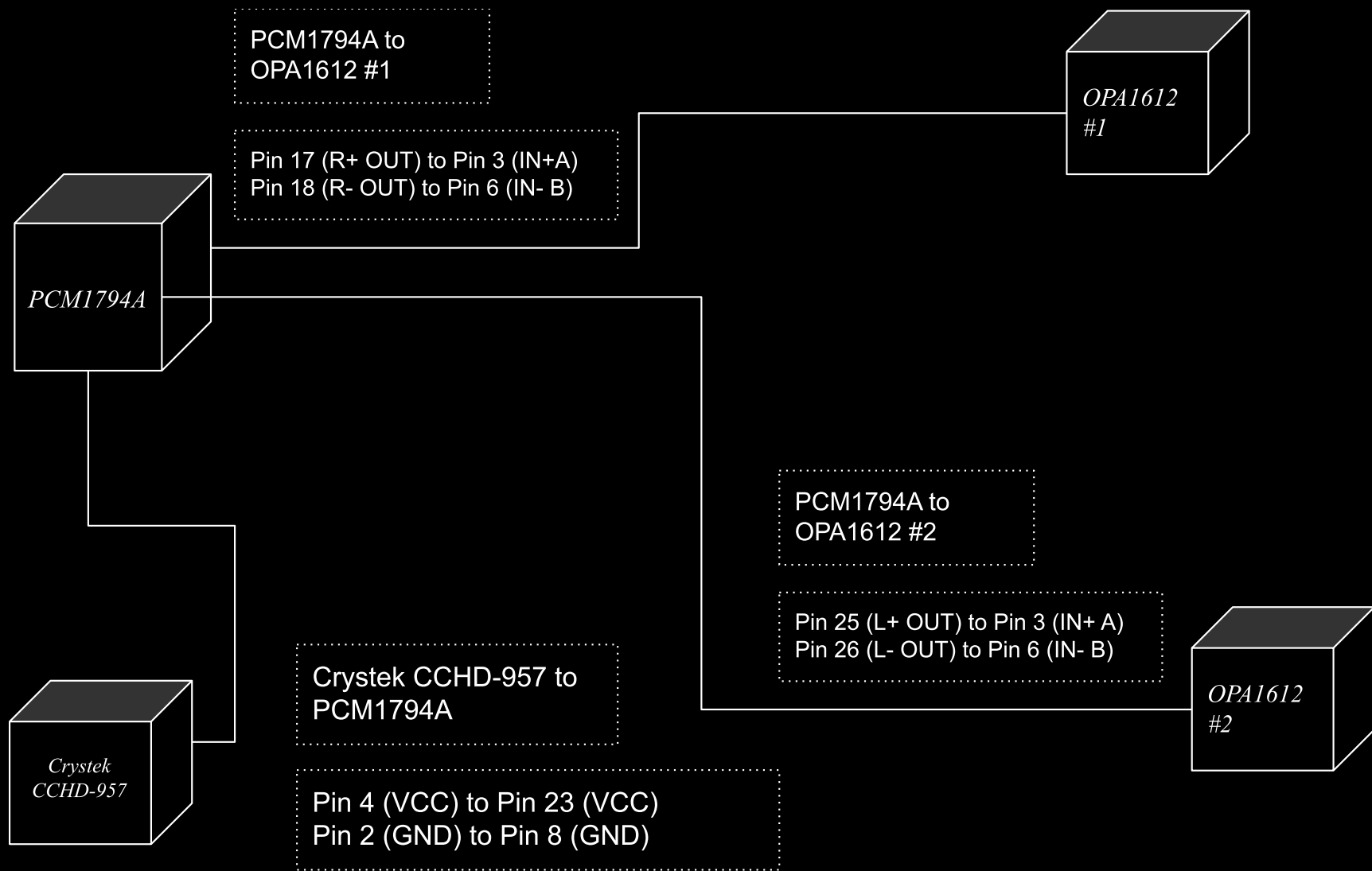
- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input

SCHEMATIC 01 - SIGNAL FLOW





SCHEMATIC 01 - PINOUT

ESP32 WROVER:

GPIO 0 (I2s Data Out) to PCM1794 (DATA)

GPIO 25 (I2S Clock/BCK) to PCM1794 (BCK)

GPIO 26 (I2S LRCLK) to PCM1794 (LRCK)

Crystek CCHD-957:

Pin 4 (VCC) to PCM1794A (Pin 23/VCC)

Pin 2 (GND) to PCM1794A (Pin 8/GND)

Pin 3 (OUT) to PCM1794 (SCK)

PCM1794A:

Pin 7(SCK) to Crystek CCHD-957 (OUT)

Pin 17 (R+ OUT) to OPA1612 #1 Pin 3 (IN+A)

Pin 18 (R- OUT) to OPA1612 Pin 6 (IN- B)

Pin 25 (L+ OUT) to OPA1612 #2 Pin 3 (IN+ A)

Pin 26 (L- OUT) to OPA1612 #2 Pin 6 (IN-B)

OPA1612 #1 (Left):

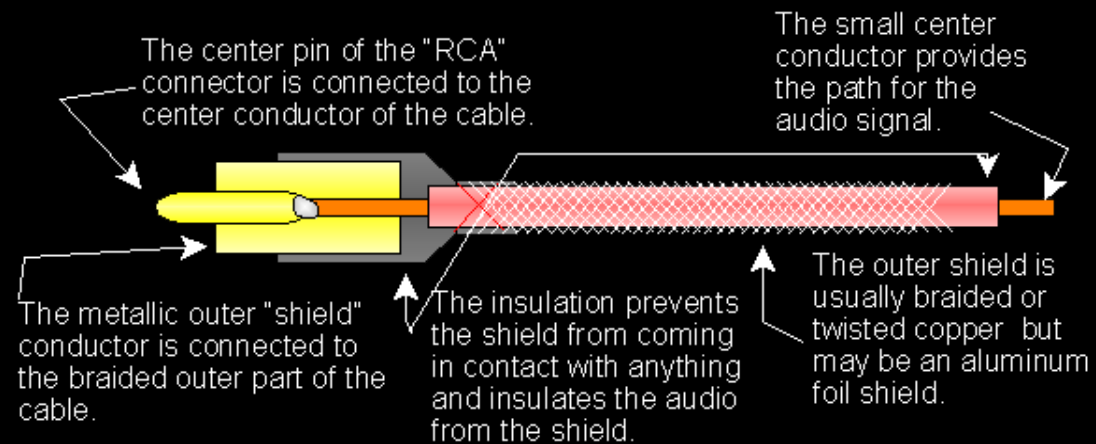
Pin 1 (OUT A) to TS Tip left to RCA

OPA1612 #2 (Right):

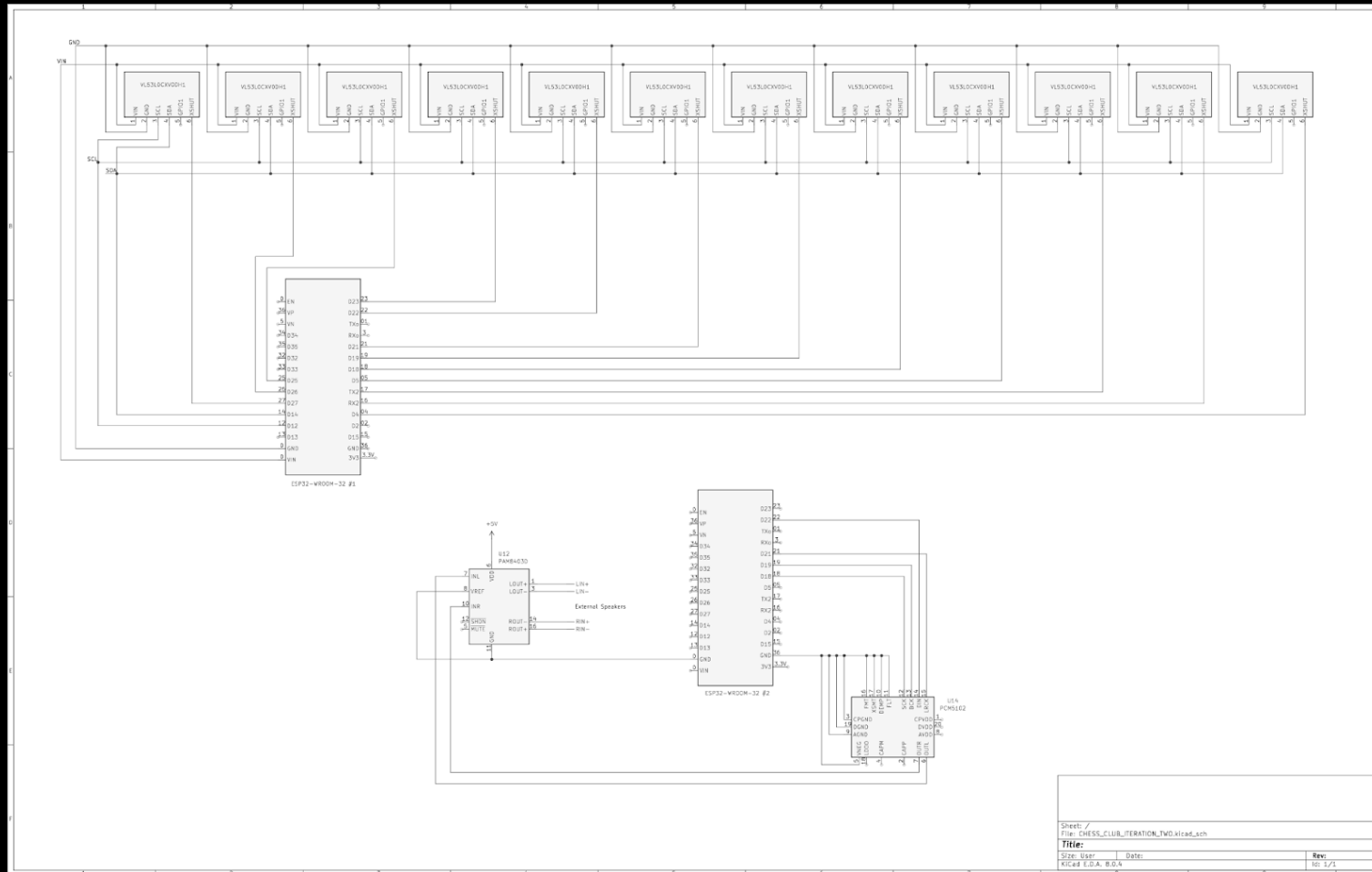
Pin 1 (OUT A) to TS Tip right to RCA

component	OUT pin	IN pin	specification
ESP32 WROVER	GPIO 0 (I2S Data Out)	PCM1794A (DATA)	I2S data output from ESP32 to DAC. Sends waveform data.
	GPIO 25 (I2S Clock / BCK)	PCM1794A (BCK)	I2S bit clock (SCK) from ESP32 to DAC.
	GPIO 26 (I2S LRCLK)	PCM1794A (LRCK)	I2S Left/Right clock (LRCLK) from ESP32 to DAC.
Crystek CCHD-957 (Oscillator)	Pin 4 (VCC)	PCM1794A (Pin 23 or VCC)	Power supply for both DAC and oscillator.
	Pin 2 (GND)	PCM1794A (Pin 8 or GND)	Ground connection.
	Pin 3 (OUT)	PCM1794A (SCK)	Master clock (MCLK) output to DAC.
PCM1794A (DAC)	Pin 7(SCK)	Crystek CCHD-957 (OUT)	Master clock (MCLK) from oscillator to DAC.
	Pin 17 (R+ OUT)	OPA1612 #1 (IN+ A) Pin 3	Analog audio output to non-inverting input
	Pin 18 (R- OUT)	OPA1612 #1 (IN- B) Pin 6	Analog audio output to the inverting input of I/V
	Pin 25 (L+ OUT)	OPA1612 #2 (IN+ A) Pin 3	
	Pin 26 (L- OUT)	OPA1612 #2 (IN- B) Pin 6	
OPA1612 (Left)	Pin 1 (OUTA)	TS Tip Left to RCA	Positive line-level output (RCA) to sub RCA input.
OPA1612 (Right)	Pin 1 (OUTA)	TS Tip Right to RCA	Positive line-level output (RCA) to sub RCA input.
RCA Subwoofer	Signal In + (Pos)	TS Tip Left to RCA	+ line-level signal input (RCA) to sub from I/V
	Signal In - (Neg)	TS Tip Right to RCA	- line-level signal input (RCA) to sub from I/V
RCA Subwoofer Output	Signal Out + (Pos)	Monitor Speakers (Signal In +)	RCA output to active monitor speakers +
	Signal Out - (Neg)	Monitor Speakers (Signal In -)	RCA output to active monitor speakers -
Active Monitor Speakers	RCA Signal In + (Pos)	RCA Subwoofer (Signal Out +)	RCA input from subwoofer output to monitor
	RCA Signal In - (Neg)	RCA Subwoofer (Signal Out -)	RCA input from subwoofer output to monitor

Notes: A transformer may be necessary in addition to shielded RCA cables



ARCHIVE



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

Preliminary Schematic Overview:

12 VL53LOCX proximity sensors connected to one ESP32.

PCM5102 DAC > ESP32 > PAM84030 Audio Driver > R/L External Speakers.

Note for modifications in the next version:

2 dedicated ESP32s for processing sensor data

VL53L4CX substitute

PRELIMINARY SCHEMATIC OVERVIEW

** power*

- 3.3V power supply
 - VCC + GND
- 20 power (VDD) terminals to 3.3V
 - 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speaker

** inputs*

- *sensors*
 - VIN to 3.3V
 - GND to GND
 - SCL to ESP32 SCL
 - SDA to ESP32 SDA
 - GPIO (none)
 - XSHUT to unique GPIO on ESP32
- *microphone*

** processors*

- ESP32s
- amplifiers
- DAC (*Digital-to-Analog Converter*)

*** outputs**

- speakers
- subwoofer
- oled display